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A Modal Radar Cross Section of Thin-Wire Targets via the Singularity Expansion Method

M. A. Richards, T. H. Shumpert, and L. S. Riggs

Abstract—**A modal radar cross section (RCS) of arbitrary wire scatterers is constructed in terms of SEM parameters. Numerical results are presented for both straight and L-shaped wire targets and are compared to computations performed in the frequency domain using the method of moments.**

I. Introduction

The calculation of the radar cross section (RCS) of various targets has been of considerable interest for the past several years, and during this time, much progress has been made in this area. New techniques have been developed [1], and some more traditional techniques have been extended and improved [2], [3], to allow an ever-widening class of problems to be solved.

Although the ability to solve complex RCS problems has increased at an astounding rate, a fundamental understanding of the physics has proceeded at a somewhat slower pace. The purpose of this communication is to apply the SEM [4], a method that has in some cases led to an increased understanding of transient scattering, to the calculation and analysis of RCS.

In this communication, first, the RCS is defined in terms of a general scattering parameter. Then, using the SEM, this parameter is written as a singularity series over the complex resonant frequencies of the scatterer, and the amplitude of each term in the series is shown to be proportional to the product of two coupling coefficients. For a given mode, one coupling coefficient governs how much of the incident field couples to the structure, while the other governs how much the structure couples to the reradiated field.

Following the analytical development, numerical results are presented for the modal RCS of straight and L-shaped, thin-wire targets. Two different coupling coefficients are used in the calculations. These are, the usual class-one form where the coefficients are evaluated at the pole locations, and a real-frequency, class-two form where the coefficients are evaluated along the $j\omega$ axis. Also, the effect of truncating the series expansion is discussed. All results are compared to a frequency-domain, method-of-moments calculation of the RCS, and, in general, agreement is excellent.

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II. Derivation of the Modal RCS

Let us define the bistatic scattering cross section, $\sigma_{\xi\gamma}$, as an equivalent area that receives that amount of power from the incident field, $E_\gamma^i(\theta^i, \phi^i)$, so as to produce the scattered field, $E_\xi^s(\theta^s, \phi^s)$, by isotropic radiation. In equation form this cross section can be written as

$$\sigma_{\xi\gamma} = \lim_{kr\to\infty} 4\pi r^2 \frac{|E_\xi^s(\theta^s, \phi^s)|^2}{|E_\gamma^i(\theta^i, \phi^i)|^2} \tag{1}$$

where both ξ and γ can be θ or ϕ, in a typical spherical coordinate system representation, for a total of four possible scattering cross sections. Notice that the usual limiting procedure associated with the definition of the RCS has been modified slightly in anticipation of the wide-band analysis that follows.

In terms of a scattering matrix, the scattered field components can be written as

$$\begin{bmatrix} E_\theta^s \\ E_\phi^s \end{bmatrix} = \begin{bmatrix} S_{\theta\theta} & S_{\theta\phi} \\ S_{\phi\theta} & S_{\phi\phi} \end{bmatrix} \begin{bmatrix} E_\theta^i \\ E_\phi^i \end{bmatrix} \tag{2}$$

where, for example, $S_{\theta\phi}$ is the far-field scattering parameter that relates the θ component of the scattered field to the ϕ component of the incident field. From (2), one can see that the appropriate definition of this parameter is

$$S_{\theta\phi} = \left.\frac{E_\theta^s}{E_\phi^i}\right|_{E_\theta^i = 0}. \tag{3}$$

Instead of considering each scattering parameter explicitly, it is convenient to derive the ξ component of the scattered field due to an incident field polarized in the γ direction, thus obtaining a general parameter $S_{\xi\gamma}$. Using this general scattering parameter, the RCS as defined in (1) becomes

$$\sigma_{\xi\gamma} = \lim_{kr\to\infty} 4\pi r^2 |S_{\xi\gamma}|^2. \tag{4}$$

The first step toward this goal of determining the general scattering parameter is to derive the current due to a γ-polarized incident field. The SEM form of the current can be found from

$$\bar{V}(s) = \bar{\bar{Z}}(s)\bar{I}(s) \tag{5}$$

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where

$\bar{V}(s)$ = voltage excitation vector
$\bar{\bar{Z}}(s)$ = system impedance matrix
$\bar{I}(s)$ = current response vector

Note that single- and double-barred quantities represent column vectors and square matrices, respectively. Equation (5) can be rewritten as

$$\bar{I}(s) = \bar{\bar{Z}}^{-1}(s)\bar{V}(s) = \sum_{\alpha} \frac{\bar{\bar{R}}_{\alpha}}{s - s_{\alpha}} \bar{V}(s) \tag{6}$$

where s_{α} and $\bar{\bar{R}}_{\alpha}$ are the αth system singularity and residue matrix, respectively. When the electric field integral equation (EFIE) is used to determine the system impedance matrix for a thin-wire geometry [5], the residue matrix is symmetric and can be written as

$$\bar{\bar{R}}_{\alpha} = \beta_{\alpha} \bar{M}_{\alpha} \bar{M}_{\alpha}^{t} \tag{7}$$

where $\bar{M}_{\alpha}$ is the αth mode vector and β_{α} is a complex proportionality constant adjusted so that the largest element in the mode vector is real and unity. Substituting (7) into (6) yields

$$\bar{I}(s) = \sum_{\alpha} \eta_{\alpha,\gamma}^{c}(s) \frac{\bar{M}_{\alpha}}{s - s_{\alpha}} \tag{8}$$

where $\eta_{\alpha,\gamma}^{c}(s)$ is the current coupling coefficient for the αth pole due to a γ-polarized incident field and is given by

$$\eta_{\alpha,\gamma}^{c}(s) = \beta_{\alpha} \bar{M}_{\alpha}^{t} \bar{V}(s) = \beta_{\alpha} \sum_{k=1}^{N} M_{\alpha}(k)(\hat{a}_{\gamma} \cdot \Delta \mathbf{l}_{k}) e^{s(r_{k}' \cdot \hat{r}_{t})/c} \tag{9}$$

where impulse excitation is assumed for the incident plane wave and

$\hat{r}_t$ unit vector pointing toward the transmitter
$\mathbf{r}_k'$ vector from the origin to the center of the kth wire segment
$M_{\alpha}(k)$ kth entry in αth mode vector
$\Delta \mathbf{l}_k$ vector length of kth current segment.

Now that the SEM form of the current has been derived, the ξ component of the scattered field can be found. The far-zone scattered field can be found by summing the field due to each zone on the structure as

$$\mathbf{E}(s) = \sum_{k=1}^{N} \Delta \mathbf{E}_{k}(s) \tag{10}$$

where $\Delta \mathbf{E}_k(s)$ is the far scattered field due to the kth zone of the wire structure and is given by

$$\Delta \mathbf{E}_{k}(s) = \frac{s\eta e^{-sr/c} e^{s(\mathbf{r}_{k}' \cdot \hat{r}_{f})/c}}{4\pi rc} I_{k}(s)[(\Delta \mathbf{l}_{k} \times \hat{r}_{f}) \times \hat{r}_{f}] \tag{11}$$

where

$\hat{r}_f$ unit vector pointing toward field point

and the assumption has been made that the distance from the structure to the observation point is much larger than the wavelength at the lowest frequency of interest. In the language of antenna theory, (11) is essentially a generalization of the far-zone field radiated from an infinitesimal (constant current) dipole. Notice that since the Laplace-domain field is directly proportional to s, the time-domain field, although not computed here, is directly proportional to the first time-derivative of the current on the wire, as usual. Furthermore, notice the direction of the field. It is that direction one would obtain from crossing the magnetic field due to the Biot–Savart Law with the unit vector $\hat{r}_f$.

To determine the ξ component of the scattered field, the total field is dotted with a unit vector in the ξ direction to yield

$$E_{\xi}(s) = \hat{a}_{\xi} \cdot \mathbf{E}(s) = \sum_{k=1}^{N} \hat{a}_{\xi} \cdot \Delta \mathbf{E}_{k}(s). \tag{12}$$

Upon examining the necessary dot product, one finds

$$\hat{a}_{\xi} \cdot [(\Delta \mathbf{l}_{k} \times \hat{r}_{f}) \times \hat{r}_{f}] = (\hat{a}_{\xi} \cdot \hat{r}_{f})(\Delta l_{k} \cdot \hat{r}_{f}) - (\hat{a}_{\xi} \cdot \Delta \mathbf{l}_{k})(\hat{r}_{f} \cdot \hat{r}_{f}) \tag{13}$$

which simply reduces to

$$\hat{a}_{\xi} \cdot [(\Delta \mathbf{l}_{k} \times \hat{r}_{f}) \times \hat{r}_{f}] = -(\hat{a}_{\xi} \cdot \Delta \mathbf{l}_{k}) \tag{14}$$

if the scattered field component of interest, ξ, is chosen to be either θ of ϕ. Since the far scattered field can always be decomposed into θ and ϕ components, this condition places no significant restriction on this development. Using (8), (11), and (14), (12) becomes

$$E_{\xi}(s) = -\sum_{k=1}^{N} \frac{s\eta e^{-sr/c} e^{s(\mathbf{r}_{k}' \cdot \hat{r}_{f})/c}}{4\pi rc} (\hat{a}_{\xi} \cdot \Delta \mathbf{l}_{k}) \cdot \sum_{\alpha} \eta_{\alpha,\gamma}^{c}(s) \frac{M_{\alpha}(k)}{s - s_{\alpha}} \tag{15}$$

which upon interchanging the order of the summations can be written as

$$E_{\xi}(s) = -\frac{s\mu}{4\pi r} e^{-sr/c} \sum_{\alpha} \frac{\eta_{\alpha,\gamma}^{c}(s)}{s - s_{\alpha}} \cdot \sum_{k=1}^{N} M_{\alpha}(k)(\hat{a}_{\xi} \cdot \Delta \mathbf{l}_{k}) e^{s(\mathbf{r}_{k}' \cdot \hat{r}_{f})/c} \tag{16}$$

where the summation on k is to be repeated for every term in the summation on α.

Notice that the latter summation in (16) takes the same form as (9) and henceforth will be termed the reradiation coefficient and defined as

$$\eta_{\alpha,\xi}^{r}(s) = \sum_{k=1}^{N} M_{\alpha}(k)(\hat{a}_{\xi} \cdot \Delta \mathbf{l}_{k}) e^{s(\mathbf{r}_{k}' \cdot \hat{r}_{f})/c}. \tag{17}$$

Furthermore, observe that the reradiation coefficient depends on the polarization and scatter direction of the scattered field, whereas the current coupling coefficient depends on the polarization and direction of incidence of the incident field. Upon defining the scattered field coefficient as the product of the current coupling coefficient and the reradiation coefficient one obtains

$$\eta_{\alpha,\xi\gamma}^{sf}(s) = \eta_{\alpha,\gamma}^{c}(s) \eta_{\alpha,\xi}^{r}(s) \tag{18}$$

which allows the scattered field to be written as

$$E_{\xi}(s) = -\frac{s\mu}{4\pi r} e^{-sr/c} \sum_{\alpha} \frac{\eta_{\alpha,\xi\gamma}^{sf}(s)}{s - s_{\alpha}}. \tag{19}$$

The scattered field coefficient as shown above is analogous to the general term in the far coupling dyad defined by Baum [6] for scattering from a general reciprocal structure. Some very important information can be extracted from (18) and (19). Namely, that the far-zone scattered field can be written as a weighted sum of terms, where the amplitude of each term is

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proportional to the product of two coupling coefficients. It is intuitively pleasing that the scattered field can be described by terms that first, characterize the coupling of the incident field to the structure, and then, the recoupling of the structure to the far field. Furthermore, notice that (18) is an interesting example of reciprocity. That is, if the transmitter and receiver are interchanged, the scattered field coefficient remains unchanged. Therefore the scattered field satisfies reciprocity on a modal basis and ensures the reciprocal nature of the total field will be maintained.

The general scattering parameter $S_{\xi\gamma}$ was described earlier as that number that relates the ξ component of the scattered field to the γ component of the incident field. When a unit plane wave polarized solely in the γ direction is assumed incident on the body, this parameter is equivalent to the ξ component of the scattered field. We therefore have

$$S_{\xi\gamma} = -\frac{s\mu}{4\pi r} e^{-sr/c} \sum_{\alpha} \frac{\eta^{sf}_{\alpha,\xi\gamma}(s)}{s - s_\alpha}. \tag{20}$$

Finally, using (4) and (20), the RCS can be written in terms of the complex frequency variable s as

$$\sigma_{\xi\gamma} = \frac{\mu^2}{4\pi} \left| \sum_{\alpha} \frac{s\eta^{sf}_{\alpha,\xi\gamma}(s)}{s - s_\alpha} \right|^2. \tag{21}$$

Since the RCS as shown in (21) is described completely in terms of SEM quantities, it will be termed the modal RCS in reference to the mode vectors used in calculating the scattered field coefficients.

When calculating the real-frequency RCS, the above equation can be evaluated along the $j\omega$ axis to yield

$$\sigma_{\xi\gamma} = \frac{(\omega\mu)^2}{4\pi} \left| \sum_{\alpha} \frac{\eta^{sf}_{\alpha,\xi\gamma}(j\omega)}{j\omega - s_\alpha} \right|^2. \tag{22}$$

This equation will be used to obtain most of the numerical results to be presented in the next section and will be referred to as the class-two, modal RCS, since it employs the frequency-dependent class-two form of the scattered field coefficient. Another interesting result can be obtained from (21) by evaluating the scattered field coefficient at the pole location to obtain

$$\sigma_{\xi\gamma} = \frac{(\omega\mu)^2}{4\pi} \left| \sum_{\alpha} \frac{\eta^{sf}_{\alpha,\xi\gamma}(s_\alpha)}{j\omega - s_\alpha} \right|^2 \tag{23}$$

which will be referred to as the class-one modal RCS. Notice that for a given pole/mode combination, the class-one coefficient is frequency independent while the class-two form of the coefficient must be evaluated for each frequency of interest. Furthermore, observe that a class-two coefficient can be evaluated at any desired complex frequency and therefore must simplify to the class-one coefficient when evaluated at the complex frequency corresponding to the pole location. As will be shown in the following section, the simplification achieved through the use of the class-one form is at the expense of the numerical accuracy of the final result.

III. Numerical Results

In this section, the RCS of both straight and L-shaped, thin-wire structures will be discussed and numerical results presented. Modal RCS results determined using both the class-two coupling coefficients and the class-one coefficients will be shown and compared to computations performed in the frequency domain using the method of moments. All frequency domain results will be taken as correct, and the performance of the modal RCS will be based upon agreement between the two methods. It will be shown that the class-two modal RCS agrees extremely well with frequency domain results while the class-one modal RCS seems to be more sensitive to numerical errors in the SEM poles and mode vectors. This is not entirely without precedent, for Baum and Pearson [7] have already shown that for transient scattering the class-one representation was particularly sensitive to pole errors.

Consider the straight wire of length 1 m and length-to-radius (l/a) ratio of 200 shown in Fig. 1. For all of the calculations discussed below, the wire is divided into 51 moment method zones.

The monostatic, class-two, modal RCS of the straight wire has been calculated at the zenith angle $\theta = 30°$ using the first four pole pairs and eight pole pairs in the first layer, respectively. These results are shown in Fig. 2. Also shown in Fig. 2 is the RCS as determined in the frequency domain via the method of moments. For the four-pole and eight-pole pair cases, one symbol has been plotted for every fourth and sixth data point, respectively. Notice that the three curves agree very well until slightly past the peak of the fourth resonance where the curve calculated using only four pole pairs diverges from the other two curves. This is to be expected and is indicative of the general trend. Since the singularity expansion contains a finite number of terms, there is some frequency above which the modal RCS will no longer be accurate.

As an extension of the case considered above, the class-one modal RCS was also calculated with eight pole pairs in the singularity expansion. This result, along with the class-two modal RCS from Fig. 2 are shown in Fig. 3. The difference between the two results is due to error in the class-one computation. Other modal RCS calculations indicate that, in general, the class-one form does not perform as well as the class-two form. As mentioned previously, this is probably because the class-one form is sensitive to numerical errors in the calculation of the poles and mode vectors. For the remainder of this analysis, the modal RCS will be determined using class-two coupling coefficients only.

The bistatic, class-two, modal RCS of the straight wire was calculated using four and eight pole pairs in the singularity expansion. The incident and scattering angles were chosen as $\theta^i = 30°$ and $\theta^s = 60°$, respectively. The resulting curves along with the frequency domain result are shown in Fig. 4. Again, the result obtained using four pole pairs diverges from the other two curves at approximately the fourth resonant frequency of the straight wire, while the result calculated using eight pole pairs agrees well with the frequency domain computation until approximately the seventh resonant frequency of the wire. The high-frequency error in the eight-pole pair calculation can be corrected by using more terms in the singularity expansion.

The final example for the straight wire to be presented here is the bistatic, class-two, modal RCS versus θ^s, the scattering angle. In this result, eight pole pairs were used and the angle of incidence and operating frequency were chosen as 45° and 600 MHz, respectively. This result along with the frequency domain computation are shown in Fig. 5. Notice that the maximum error occurs near the specular angle of 135°. In other examples, not shown here, the specular angle was repeatedly found to be location of the maximum error, regardless of the angle of incidence. Furthermore, using more terms in the singularity expansion, including both first and second layer poles, did not, in general, significantly improve the modal RCS results near the specular angle. Although this phenomenon has not been studied

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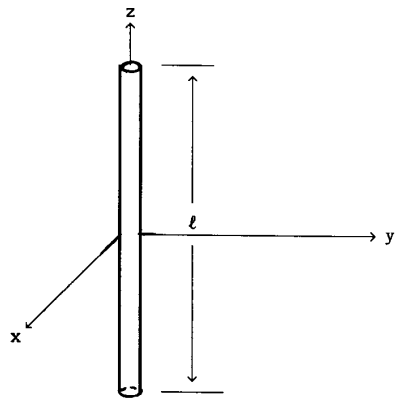

Fig. 1. Straight thin-wire scatterer of length $l = 1$ m and length-to-radius (l/a) ratio of 200.

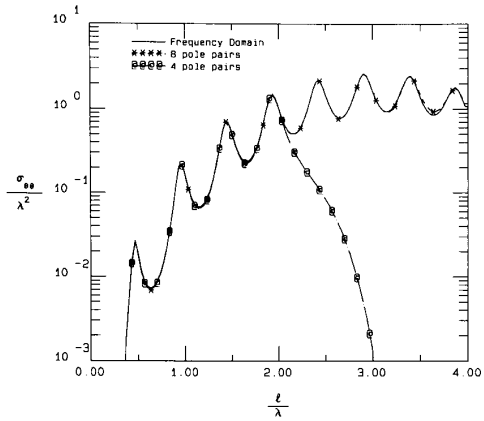

Fig. 2. Monostatic RCS of a straight wire, calculated at $\theta = 30°$ using the SEM with four and eight pole pairs and compared to a frequency domain/method of moments result.

exhaustively, it is believed that this type of inaccuracy is due to numerical errors in the poles and mode vectors.

The remaining numerical results to be presented here are for the L-shaped wire scatterer shown in Fig. 6. It has a total length of 1 m, a length-to-radius (l/a) ratio of 200, and is divided into 59 moment method zones. All modal RCS results shown below were calculated using class-two coupling coefficients and eight pole pairs in the singularity expansion.

As the first example calculation for the L-shaped wire, the co-polarized, bistatic, modal RCS was determined due to a ϕ-polarized, 600 MHz incident plane wave. The incident angles were chosen to be $(\theta^i = 60°, \phi^i = 60°)$ and the RCS was calculated in the constant azimuthal plane $\phi^s = 60°$. The resulting RCS is shown versus the zenith scattering angle θ^s in Fig. 7. Also shown in this figure is the frequency domain result for comparison purposes. The agreement between the two methods is excellent.

The final example to be considered is the cross-polarized, bistatic RCS, $\sigma_{\phi\theta}$, due to a θ-polarized, 900 MHz, incident plane wave. The incident angles were chosen as $(\theta^i = 30°, \phi^i = 30°)$ and the RCS was calculated as a function of the azimuthal scattering angle ϕ^s, with the zenith scattering angle held constant at $\theta^s = 60°$. This result, along with the frequency domain result, is shown in Fig. 8.

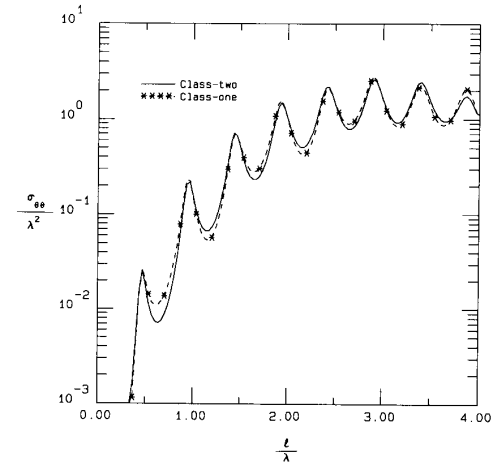

Fig. 3. Monostatic RCS of a straight wire, calculated at $\theta = 30°$ using the SEM with eight pole pairs with both class-one and class-two coupling coefficients.

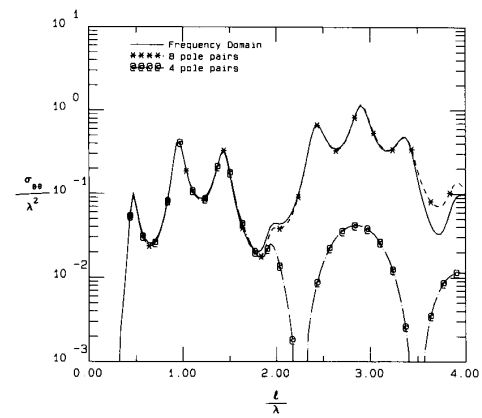

Fig. 4. Bistatic RCS of a straight wire, calculated using the SEM with four- and eight-pole pairs and compared to a frequency domain result. The angles of incidence and scattering are $\theta^i = 30°$ and $\theta^s = 60°$, respectively.

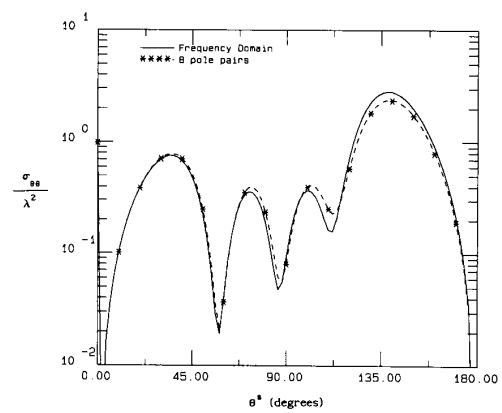

Fig. 5. Bistatic RCS of a straight wire, calculated at a frequency of 600 MHz using the SEM with eight-pole pairs and compared to a frequency domain result. The angle of incidence is $\theta^i = 45°$.

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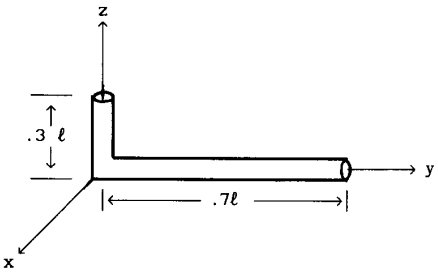

Fig. 6. L-shaped thin wire scatterer of total length $l = 1$ m with a length-to-radius (l/a) ratio of 200.

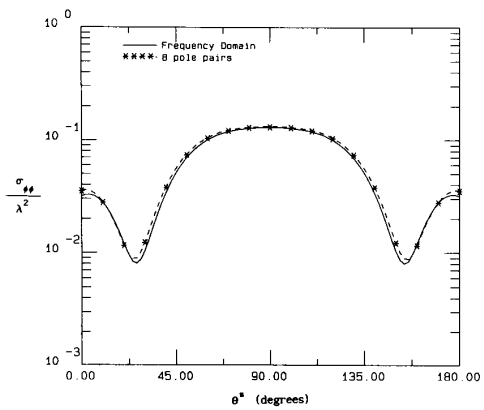

Fig. 7. Co-polarized, bistatic RCS of an L-shaped wire as a function of the zenith scattering angle, θ^s, calculated using the SEM with eight-pole pairs and compared to a frequency domain result. The excitation is a 600 MHz, ϕ, plane wave incident from ($\theta^i = 60°$, $\phi^i = 60°$), and the constant azimuthal scattering angle is $\phi^s = 60°$.

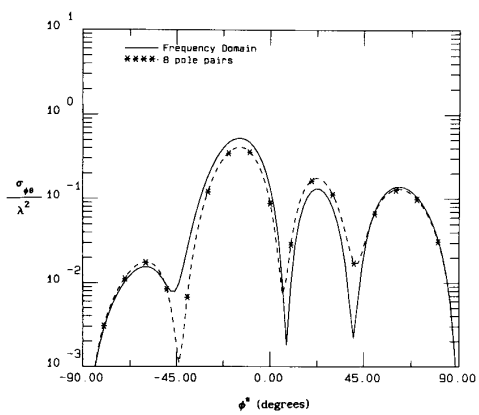

Fig. 8. Cross-polarized, bistatic RCS of an L-shaped wire as a function of the azimuthal scattering angle, ϕ^s, calculated using the SEM with eight-pole pairs and compared to a frequency domain result. The excitation is a 900 MHz, θ-polarized, plane wave incident from ($\theta^i = 30°$, $\phi^i = 30°$), and the constant zenith scattering angle is $\theta^s = 60°$.

IV. Conclusion

It has been shown that using the SEM, the RCS of arbitrary wire structures can be cast into a particularly simple form. Furthermore, the parameters in the resulting equation have significant physical importance. Numerical results have been presented to validate the method.

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Microstrip Antennas and Arrays on Chiral Substrates

David M. Pozar

Abstract—**Results are presented for isolated microstrip antennas and infinite arrays of microstrip antennas printed on chiral substrates, computed from full-wave spectral domain moment method solutions. Data for resonant length, impedance, directivity, efficiency, cross-polarization level, and scan performance are given, and compared to results obtained for a dielectric substrate of the same thickness and permittivity. It is concluded that, from the point of view of antenna characteristics, there does not seem to be any advantage to using chiral antenna substrates, while there are disadvantages in terms of increased cross-pol levels and losses due to surface wave excitation.**

Introduction

Chiral materials are isotropic, reciprocal, and exhibit optical activity, whereby the propagation constants are different for right- and left-hand circularly polarized (RHCP, LHCP) waves. A considerable amount of analytical work in recent years has been devoted to the application of chiral materials to problems in electromagnetic propagation, scattering and radiation [1]–[4], including the use of chiral materials for printed transmission lines [5] and printed antennas [6].

One of the characteristics of wave propagation in infinite chiral media is that the eigenfunctions consist of decoupled RHCP and LHCP modes. This situation is similar to that of plane waves propagating in the direction of the bias field in an infinite magnetized ferrite medium [7], although in this case the material is anisotropic and nonreciprocal. Recently it has been shown that the use of ferrite materials for microstrip antenna and array substrates can provide several novel antenna characteristics that cannot be obtained with ordinary dielectric substrates, including tunable and switchable circular polarization

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